

Assignment 2 (100 marks)

You are given a 3D domain, owned by $p_x \times p_y \times p_z$ processes. Each sub-domain size is $n_x \times n_y \times n_z$. Iterate over T steps, Dimension of Data[F] per process = $n_x * n_y * n_z$ doubles, F is the number of fields/variables.

Communication: halo exchange for d-point stencil

Computation: d-point stencil (d is input, $d < n_x$, $d < n_y$, $d < n_z$, example values of d are 7, 13, ...)

For e.g., average of E,W,N,S,F,B and itself for $d=7$, i.e. value $(P, t+1) = [\text{value}(P_W, t) + \text{value}(P_E, t) + \text{value}(P_N, t) + \text{value}(P_S, t) + \text{value}(P_F, t) + \text{value}(P_B, t) + \text{value}(P, t)] / 7$

Each process computes the count of isovalues (i.e. the count of points in the domain that would be equal to a given isovalue, refer to Lectures 17 and 18)

Root process calculates the total count of isovalues for every field across all time steps, i.e. there will be T values for field #1, T values for field #2, ..., and T values for field #F.

The initial data must be randomly generated using the value "seed" as shown below.

```
srand(seed);
for (int i=0; i<F; i++)
    for (int j=0; j<arrSize; j++)          // arrSize =  $n_x * n_y * n_z$ 
        data[i][j] = (double)rand()*(myrank+1)/(110426.0+i+j);
// you are free to use any data layout but the initialization should be done like this.
```

The input to the program is d, ppn, p_x , p_y , p_z , n_x , n_y , n_z , T, random seed, F, isovalue

The output (by rank 0) should be **T+1 lines**, the first T lines contain F numbers, followed by the time:

```
<count_field1> ... <count_fieldF> // for time step 1
...
<count_field1> ... <count_fieldF> // for time step T
<time> [put a new line at the end]
```

Final configurations for the report: (there are a total of 8 configurations)

for execution in 1 to 5 // repeat each configuration 5 times

P (number of processes) in 32, 48, 64, 96 (4 configurations)

if $P = 32$ then $p_x = 4$, $p_y = 4$, $p_z = 2$, $ppn = 32$

if $P = 48$ then $p_x = 6$, $p_y = 4$, $p_z = 2$, $ppn = 48$

if $P = 64$ then $p_x = 4$, $p_y = 4$, $p_z = 4$, $ppn = 32$

if $P = 96$ then $p_x = 6$, $p_y = 4$, $p_z = 4$, $ppn = 48$

$n_x = n_y = n_z = 120$, $n_x = n_y = n_z = 240$ (2 configurations)
 $d = 7$
 $ppn =$ as per above // For PARAM Rudra (note: this will be set as per the test environment)
 $T = 5$
 $seed = 1000$
 $F = 2$
 $isovalue = 500$

You **must only** use the MPI functions discussed in class. You may optimize your code using concepts learnt in class. Submit your optimized code only.

Due date: 6 PM on 12-04-2026 (Hard deadline)

Late submission (-10 penalty): 6 PM on 13-04-2026

Assignment 2 doubts: <https://forms.gle/EQuyHDQCsqdSp8NP7>

FAQ

Q: Do we double count the boundary isovalue counts?

A: Yes, as discussed in class (after Lecture 19).

Q: Are we searching for the same isovalue (500) for different variables? Variables may have different ranges.

A: Yes, in this assignment we are searching for the same isovalue (input argument) for all variables.

Q: In Assignment 1 we were not allowed to use `MPI_Isend` and `MPI_Irecv`. Does the same restriction apply to Assignment 2 as well, or can we use non-blocking communication functions for the halo exchange?

A: Please read the document "You **must only** use the MPI functions discussed in class." `MPI_Isend` and `MPI_Irecv` have been covered, hence can be used.

Q: Can we assume for d point stencil computation, the data needed for halo is available with immediate neighboring ranks or do we need to consider the scenario where the data spans multiple neighboring ranks on either direction. For example, do we need to take care of the scenario where each process has $2 \times 2 \times 2$ data and we have to compute 19 point stencil.

A: Yes, that is why the extent of d was given as $d < n_x$, $d < n_y$, $d < n_z$